Elasticity: the material answers back

Moduli are material properties; stiffness is geometry. Everything else is bookkeeping.

What elasticity is

$$\sigma = \frac{F_\perp}{A}, \qquad \varepsilon = \frac{\Delta L}{L}$$

$$\gamma = \text{shear angle in radians}, \qquad \frac{\Delta V}{V}$$

A solid resists a change of shape and returns when the load is removed. Stress is internal force per unit area on a named cut plane; strain is the dimensionless geometric response. Four modes: tension, compression, volume change and shear.

WHY / Every atom in the cross-section shares the load, so the same force on half the area is twice the stress. A bond breaks at a stress, never at a force, which is why allowable loads scale with area.

WATCH / Changing the length changes the extension but not the stress in a uniform wire under an end load. If your answer needs L to decide closeness to breaking, you have used the wrong variable.

The four bookkeeping pairs

$$Y = \frac{\sigma_n}{\varepsilon}, \qquad B = -\frac{\Delta p}{\Delta V/V}, \qquad G = \frac{\tau}{\gamma}$$

$$[Y] = [B] = [G] = \text{Pa} = \text{N m}^{-2}$$

Tensile/compressive tests give Y. A cube squeezed on all faces gives B. A block sheared by opposing face forces gives G. The minus sign in B keeps it positive: pressure up means volume down. Strain is dimensionless, so every modulus is a pressure.

WHY / The three moduli are not three materials: they are three questions asked of one material. For an isotropic solid two numbers fix the rest, which is what the Poisson relations do.

WATCH / Modulus is independent of size and shape only inside the linear range and at fixed temperature. Stiffness k = YA/L is not a material property; it carries the geometry.

Reading a stress-strain curve

$$\sigma_{\text{working}} = \frac{\sigma_{\text{yield}}}{n}, \qquad n \approx 1.5 \text{ to } 3$$

In order: proportional limit (Hooke dies), elastic limit (recovery dies), yield (strain runs at nearly constant stress), ultimate strength (necking starts), fracture. Ductile metals draw the whole story; brittle solids such as glass snap just past the elastic limit with no warning.

WHY / The curve is design information. Working stress is a fraction of yield, never of the ultimate, because the ultimate is reached only after the member has already deformed permanently.

WATCH / The area under the elastic part is recoverable energy per volume; the total area out to fracture is toughness. Do not read the strain axis as a strain rate, and do not compare Y of two materials from the slopes of F against extension.

The elastic region is tiny

$$\varepsilon_{\text{elastic}} \lesssim 10^{-3} \text{ for metals}$$

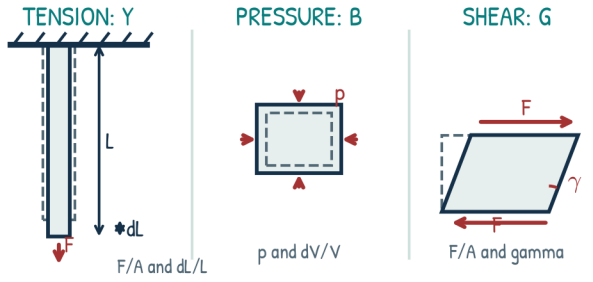
$$\sigma = Y\varepsilon \;\Rightarrow\; \varepsilon = \frac{2.5 \times 10^{8}}{2.0 \times 10^{11}} = 1.25 \times 10^{-3}$$

Mild steel yields at about 250 MPa with Y = 200 GPa, so the straight part of its curve ends near a strain of 0.1 %. The whole elastic story happens before the material has changed length by one part in a thousand.

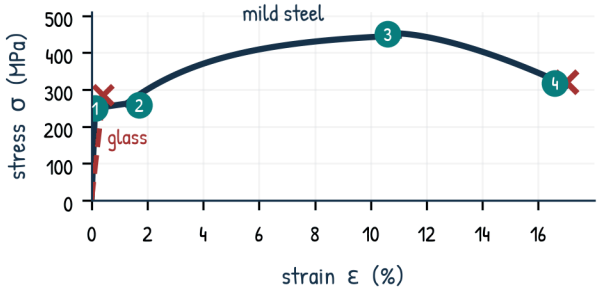
WHY / Small strain is why the original geometry may be used in every formula: A and L do not change enough to matter before the law stops applying.

WATCH / Beyond yield the same symbols mean different physics. An answer that uses Y at a strain of 0.05 is arithmetically fine and physically meaningless.

The three deformation modes and the modulus each one defines



1 proportional limit, 2 yield, 3 ultimate, 4 fracture; steel vs glass



REMEMBER / Stress is force democratised: failure is set by force per area, not by force.

Y, B, G and what Poisson ties together

Two constants describe an isotropic linear solid; the other two follow.

Poisson ratio defined

$$\nu = -\frac{\varepsilon_{\text{lat}}}{\varepsilon_{\text{long}}}$$

$$\frac{\Delta V}{V} = \varepsilon(1-2\nu)$$

A wire stretched by strain e thins laterally by -nu e. The definition carries a minus so ordinary materials give nu > 0. Volume change under uniaxial tension is (1-2nu)e: at nu = 0.5 the volume is unchanged, and at nu = 0 the wire keeps its cross-section.

WHY / The lateral contraction is a real strain that must be compatible with everything attached to the wire. It is why a constrained specimen carries a different effective stiffness from a free one.

WATCH / nu corrects the lateral strain only. The length change of a uniaxial test is dL = FL/AY, with no Poisson factor. Do not smuggle (1+nu) into dL.

Derive Y = 3B(1 - 2nu)

$$\varepsilon_1 = -\frac{p}{Y} + \frac{\nu p}{Y} + \frac{\nu p}{Y} = -\frac{p}{Y}(1-2\nu)$$

$$\frac{\Delta V}{V} = 3\varepsilon_1 = -\frac{3p(1-2\nu)}{Y} = -\frac{p}{B}$$

Apply hydrostatic pressure p. Each of the three principal strains gets -p/Y from its own axis and +nu p/Y from each of the other two. Add the three strains for the volume strain and equate to the definition of B.

WHY / Only two elastic constants are independent. B measures resistance to volume change, G resistance to shape change at fixed volume, and this derivation is the exact split between them.

WATCH / Using deviatoric stress with a bulk modulus is the classic error. B responds only to the mean stress p; a pure shear stress has zero mean and produces no volume change in a linear isotropic solid.

Derive Y = 2G(1 + nu)

$$\tau\ \text{shear} = \tau\ \text{tension} / -\tau\ \text{compression}$$

$$\frac{\tau}{2G} = \frac{\tau}{Y}(1+\nu) \ \Rightarrow\ Y = 2G(1+\nu)$$

Pure shear rotated by 45 degrees is tension on one diagonal and equal compression on the other. The diagonal strain from those normal stresses is (tau/Y)(1+nu); the geometry of a sheared square gives the same quantity as gamma/2, and gamma = tau/G. Equate.

WHY / This is why a torsion test can measure a tensile modulus. The two loadings are the same physical state seen in different coordinates, and superposition of principal strains is the whole proof.

WATCH / The rotation is 45 degrees for every isotropic material, but the equality of the two principal stresses holds only for pure shear, not for a general stress state.

Why -1 < nu < 0.5

$$B = \frac{Y}{3(1-2\nu)} > 0, \qquad G = \frac{Y}{2(1+\nu)} > 0$$

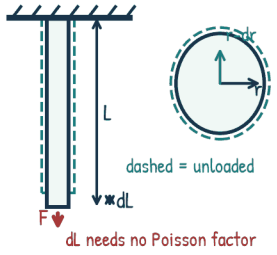
$$\nu \to \tfrac{1}{2}:\ B \to \infty; \qquad \nu \to -1:\ G \to \infty$$

Invert the two relations. A stable solid needs B > 0 and G > 0, which forces nu < 0.5 and nu > -1. Rubber sits near 0.5 (nearly incompressible); cork near 0 (it compresses without bulging); auxetic foams have nu < 0.

WHY / The bounds are thermodynamic: a negative modulus would let the solid lower its energy without limit by deforming, so the state is unstable, not merely unusual.

WATCH / nu = 0.5 is a limit, not a value: it needs an infinite bulk modulus. Real rubber is 0.499, which still gives B about a thousand times G.

Poisson contraction: lateral strain is -nu times longitudinal strain

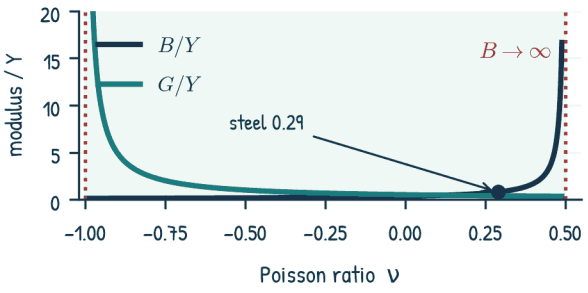


$$\nu = -\varepsilon_{\text{lat}}/\varepsilon_{\text{long}}$$

$$\Delta r = -\nu r\, \Delta L/L$$

$$\Delta V/V = (1-2\nu)\varepsilon$$

Stable solids need -1 < nu < 0.5; B diverges as nu tends to 0.5



REMEMBER / Any two of Y, B, G and nu fix all four; the bounds are energy demands, not conventions.

The extension family: rods as springs

Compute k = YA/L once, then series and parallel decide everything else.

Rod as a spring

$$\Delta L = \frac{FL}{AY} \iff F = \frac{YA}{L}\Delta L = k\Delta L$$

$$k = \frac{YA}{L}$$

Y is the material, A/L is the geometry, and their product is the spring constant. Doubling the length halves k (two springs in series); doubling the area doubles k (two springs side by side). Cutting a wire into n pieces multiplies the stiffness of each by n.

WHY / This one line converts every elasticity problem into a spring problem already solved in SHM: same k, same 1/2 k x^2 energy, same omega = sqrt(k/m).

WATCH / A = pi d^2/4. Using pi d^2, or treating a diameter as a radius, is a factor of four and the commonest arithmetic error in the chapter.

Series: common force

$$\frac{1}{k_{\text{tot}}} = \sum_i \frac{1}{k_i}$$

$$\Delta L = F\sum_i \frac{L_i}{A_i Y_i}$$

Two rods welded end to end feel the same F; each extends by F L_i/(A_i Y_i) and the extensions add. The stress steps at the joint whenever the area changes: sigma_i = F/A_i, larger in the thinner segment.

WHY / The total extension depends on the sum of compliances. Engineers write it as a sum of L/AY because that is what a spreadsheet can accumulate along a real member.

WATCH / Do not assume equal stress in series. Equal force with unequal area is unequal stress, and the thin side is the one that fails first.

Parallel: common extension

$$k_{\text{tot}} = \sum_i k_i, \qquad \frac{F_1}{F_2} = \frac{k_1}{k_2}$$

$$k_s = 2.0\times10^7:\ F_s = 952\ \text{N};\ k_c = 2.2\times10^7:\ F_c = 1048\ \text{N}$$

Two wires hanging from one rigid bar extend together. The load divides in proportion to stiffness. In the worked pair (steel 1 cm^2, copper 2 cm^2, 2000 N) the copper carries the larger share although its modulus is lower, because its area more than compensates.

WHY / Compatibility writes the equation, not the force balance alone: the bar is rigid, so the two extensions are equal. That single geometric statement is the whole method.

WATCH / A rigid bar is an idealisation. If the bar can tilt, the extensions need not be equal and the problem acquires a moment balance as well.

Self-weight: integrate, do not average

$$F(x) = \rho A g x, \qquad \Delta L = \int_0^L \frac{F(x)\,dx}{AY} = \frac{\rho g L^2}{2Y}$$

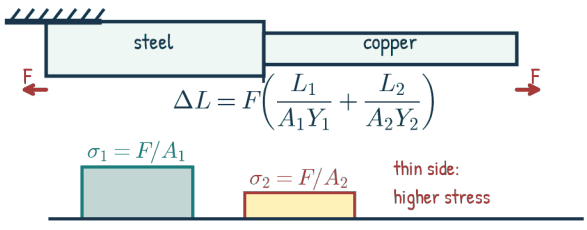
$$= \frac{(mg/2)L}{AY}$$

A hanging rod carries, at height x above its free end, only the weight below it. The load ramps from zero to mg, so the extension equals that of half the weight hung at the tip. A tapered rod uses the same integral with A(x) inside.

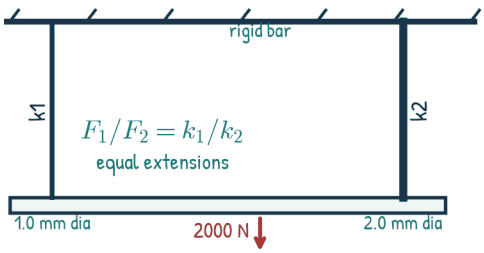
WHY / The half is the integral of x, not a convention: the same triangular-load result as the centroid of a triangle.

WATCH / Using the full weight gives exactly twice the true extension. If your answer to a self-weight question equals the end-load formula with mg, you averaged where you should have integrated.

Series composite: common force, extensions add, stress steps with 1/A



Parallel wires under a rigid bar: equal extensions, load splits as stiffness



REMEMBER / Series shares force and adds extensions; parallel shares extension and splits force as stiffness.

Composite bars and thermal stress

Two-step thermal work: expand freely, then restore the constraint.

Thermal strain and stress

$$\Delta L_{\text{free}} = \alpha L \Delta T$$

$$\sigma_{\text{th}} = Y\alpha\Delta T, \qquad F = Y\alpha\Delta T\,A$$

Heat a rod between rigid walls. Left free it would grow alpha L dT; the walls push it back by FL/AY. Net length change is zero, so FL/AY = alpha L dT. Notice L cancels: a short bolt and a kilometre of rail develop the same stress for the same temperature change.

WHY / The wall provides whatever force is needed to restore the constraint, so the stress is a reaction, not a material response to heat. Geometry enters only through the area that carries that reaction.

WATCH / The sign comes from the constraint. Heating a fully constrained rod compresses it, cooling tensions it. Writing sigma = +Y alpha dT for cooling is the standard sign error.

Gaps, partial constraint and the rail

$$\text{gap } g\text{: no stress until } \alpha L\Delta T > g$$

$$\sigma = Y\frac{\alpha L\Delta T - g}{L} = Y\left(\alpha\Delta T - \frac{g}{L}\right)$$

A rail with expansion gaps is free until the gap closes and constrained afterwards. A 1 km steel rail with alpha = 1.2e-5 sees 0.48 m of breathing over a 40 K swing, and 96 MPa of compression if welded and held. That is about 40 % of the yield of mild steel.

WHY / Real track is installed and stressed at a chosen neutral temperature so that the summer compression and winter tension are both acceptable. The design question is a constraint question.

WATCH / In a free rail a year of daily cycles does not accumulate: expansion is not a ratchet. Ask for the swing over the peak range, once.

Two-material composite bar

$$\Delta L_1 = \Delta L_2 \text{ (bonded)}$$

$$F = \frac{(\alpha_2 - \alpha_1)\Delta T}{\frac{1}{A_1Y_1} + \frac{1}{A_2Y_2}}$$

Bond brass to steel and heat the pair. Both must reach the same final length, so the more expansive metal is held in compression and the other is pulled in tension. The common force follows from the compatibility condition, exactly as two springs in series under an imposed displacement.

WHY / The two-step method generalises: write the free expansion of each part, then the elastic correction that makes the final lengths agree. The stress is whatever mechanics needs.

WATCH / If the joint yields or the bond fails, compatibility no longer holds and the force is capped by the joint strength. The elastic formula then overestimates the stress.

Shrink fits and a quarter turn

$$\varepsilon = \frac{\Delta L}{L} = \frac{p\,n}{L}, \qquad \sigma = Y\varepsilon$$

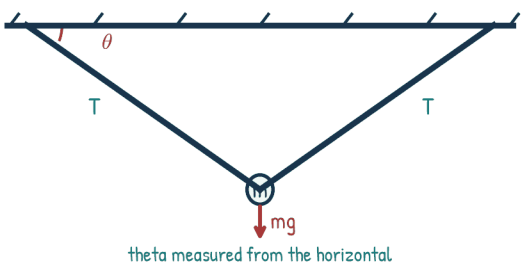
$$\text{quarter turn, } p = 1\text{ mm, } L = 200\text{ mm: } \varepsilon = 1.25\times10^{-3}$$

A brass ring is heated, slipped over a steel plug and cooled; the plug prevents the shrinkage and the ring ends in tension. A nut advanced by one quarter turn stretches the gripped length by a quarter of the pitch: a tiny angle produces a large strain because the pitch is short compared with the grip.

WHY / Torque wrenches exist because the conversion from rotation to strain is lever-like: a small advance over a short length is a large strain, and the bolt can yield before it feels tight.

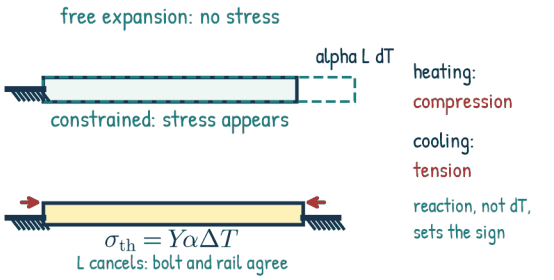
WATCH / The strain is pitch times turns divided by the gripped length, not by the bolt length. Heating a tightened bolt adds Y alpha dT if the clamped plates expand less.

Angled hanger: tension mg/(2 sin theta), not mg/2 unless theta = 90 degrees



theta measured from the horizontal

Thermal stress: expand freely, then compress back to the constraint



REMEMBER / Thermal stress is created by the constraint, not by the temperature.

Elastic energy, sudden loads and drops

At the lowest point the kinetic energy is zero, so energy closes the problem.

Stored energy

$$U = \tfrac{1}{2}F\Delta L = \tfrac{1}{2}k\Delta L^{2}$$

$$u = \tfrac{1}{2}\sigma\varepsilon = \frac{\sigma^{2}}{2Y} = \frac{Y\varepsilon^{2}}{2}$$

Stretching quasi-statically means the force grows from zero with the extension, so the work is the area of a triangle, not a rectangle. Per unit volume the same statement reads u = half stress times strain. In shear and in volume change the pattern repeats with G and B.

WHY / The energy is quadratic in the load, so doubling the load quadruples the stored energy. This is why a small overload does disproportionate damage in a spring.

WATCH / The half is not optional. U = F dL is the work of a constant force, which is not how a spring loads.

Sudden application: the factor 2

$$mg\Delta = \tfrac{1}{2}k\Delta^{2} \;\Rightarrow\; \Delta = \frac{2mg}{k} = 2\Delta_{\mathrm{st}}$$

$$\sigma_{\mathrm{peak}} = 2\sigma_{\mathrm{st}}$$

Release a weight from the slack position instead of lowering it gently. It falls, overshoots the static extension and stops where the stored elastic energy equals the potential energy lost. That occurs at twice the static extension, hence twice the static stress.

WHY / The factor two is an energy result, not a dynamic correction factor to be memorised. It is also the h = 0 limit of the drop formula, which is the check that the two results agree.

WATCH / At the lowest point the free-body diagram does not balance: the mass is accelerating upward. Demanding F = mg there is the standard wrong method.

Drop from height h

$$mg(h+\Delta) = \tfrac{1}{2}k\Delta^{2}$$

$$\Delta = \Delta_{\mathrm{st}}\left(1+\sqrt{1+\frac{2h}{\Delta_{\mathrm{st}}}}\right), \qquad F_{\max} \approx \sqrt{2hmgk}$$

The mass descends h + delta, not h, because it keeps falling while the wire stretches. Solving the quadratic gives the peak extension; for h much greater than the static extension the peak force grows as the square root of stiffness, so a stiffer wire shocks harder.

WHY / For h much greater than the static extension the answer collapses to sqrt(2hmgk). The counter-intuitive design lesson: the peak stress, not the peak force, is what must stay under yield, and the area is in k.

WATCH / Writing mgh = half k delta^2 drops the delta in the descent and undercounts the energy by roughly h/delta_st. Name the approximation if you use it.

Rivets, bolts and double shear

$$\tau = \frac{F}{nA}, \qquad n = \text{number of shear planes}$$

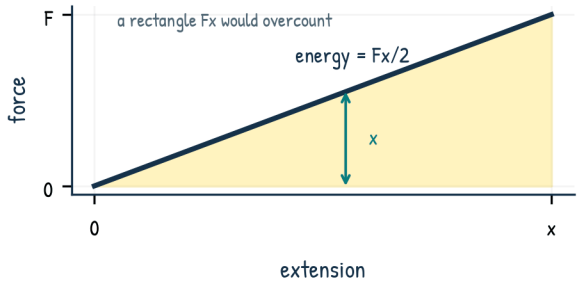
$$\tau = \frac{10^{4}}{2\pi(4.0\times10^{-3})^{2}} = 9.95\times10^{7}\ \mathrm{Pa}$$

A rivet in a lap joint is cut across one plane (single shear); a rivet between three plates is cut across two (double shear). The stress is the load divided by the total sheared area, so the plane count is the whole problem.

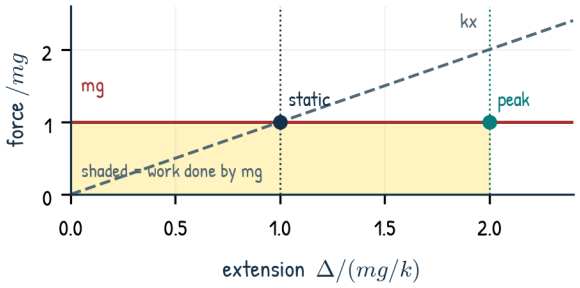
WHY / Shear failure is a cutting failure, not a stretching failure. Counting the planes is the analogue of counting the wires in a parallel bundle.

WATCH / Single shear written as double halves the stress and doubles the apparent safety. Draw the joint and count.

Linear spring: stored energy is the triangle, not the rectangle



Sudden release: energy balance gives twice the static extension



REMEMBER / Energy at the turning point, never force balance at the extreme.

Torsion, bending and section properties

Put material far from the axis and the same mass resists far more.

Torsion of a circular shaft

$$\gamma = \frac{r\theta}{L}, \qquad \tau = G\gamma = \frac{Gr\theta}{L}$$

$$T = \frac{G\theta}{L}\int_0^R 2\pi r^3 dr = \frac{G\theta}{L}\frac{\pi R^4}{2}$$

$$\theta = \frac{TL}{GJ}, \qquad J = \frac{\pi R^4}{2}$$

Twisting by theta over L shears a surface element by gamma = r theta/L, so the stress grows with radius and its torque carries an extra r: the integral is of r^3. J collects the geometry.

WHY / A torsion pendulum is this spring: C = GJ/L, and T = 2 pi sqrt(I/C) with I the inertia of the hanging body.

WATCH / J = pi R^4/2 for a solid circle, but the bending second moment is I = pi R^4/4. Swapping them is a factor-of-two error with correct dimensions, so units will not catch it.

Which section is stiffer per kilogram

$$J = \int r^2 dA, \qquad \text{core } r < 0.7R:\ 24\%\ \text{of } J,\ 49\%\ \text{of } A$$

$$\text{solid}:\ J = \frac{\pi R^4}{2}; \quad \text{thin tube}:\ J \approx 2\pi R^3 t$$

J weights area by r^2, so material near the axis contributes almost nothing to stiffness while still costing mass. Removing the core of a shaft therefore removes much area and little J, and the hollow shaft twists less for the same mass and torque.

WHY / At equal outer radius the solid shaft is stiffer; at equal mass the hollow one is. The comparison must state which quantity is held fixed, or the answer flips.

WATCH / Hollow shafts also fail locally: a thin wall buckles in shear. The stiffness gain is real, and the wall thickness has a lower limit set by a different failure mode.

Bending: neutral axis and curvature

$$\varepsilon = \frac{y}{R}, \qquad \sigma = \frac{Yy}{R}$$

$$M = \int \frac{Yy^2}{R} dA = \frac{YI}{R} \ \Rightarrow\ \frac{M}{I} = \frac{Y}{R}$$

Fibres at distance y from the unstrained neutral layer change length by y/R, so the strain and the stress grow linearly across the depth, tension on the convex side and compression on the other. The internal moment is the integral of stress times y, which defines I.

WHY / The neutral axis passes through the centroid only for a homogeneous section, because zero net axial force requires the first moment of area to vanish.

WATCH / For a composite section the axis sits at the stiffness-weighted centroid. Assuming the geometric centre mis-assigns stress to both materials.

Cantilever and the I-beam

$$\frac{d^2y}{dx^2} = \frac{M(x)}{YI}, \qquad M = F(L-x)$$

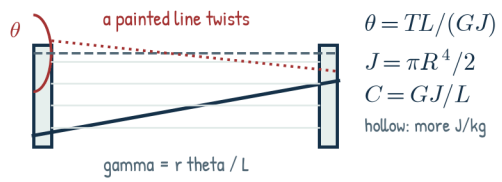
$$\delta_{\text{tip}} = \frac{FL^3}{3YI}, \qquad I_{\text{rect}} = \frac{bh^3}{12}$$

Integrating the curvature twice with zero slope and deflection at the wall gives FL^3/3YI. Depth enters I as h^3 and width as b, so turning a rectangle through 90 degrees costs a factor of four. An I-profile wins about two at equal depth and area.

WHY / Deflection scales as L^3, stress as L: a long beam fails by being too floppy, a short one by being too weak. Different checks.

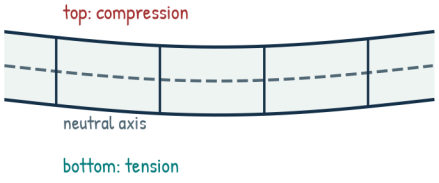
WATCH / The factor of five often quoted compares a deep I-section with a shallow rectangle and silently credits the depth.

Torsion: shear strain grows with radius; J = pi R^4 / 2



$\theta = TL/(GJ)$
$J = \pi R^4/2$
$C = GJ/L$
hollow: more J/kg

Bending: strain grows with distance y from the neutral axis



$M = YI/R$
$I = \int y^2 dA$
$\sigma = Yy/R$
y from the axis

REMEMBER / Section properties are geometry: compute I and J before inserting any material.

Yield, fatigue and choosing a material

Strength, stiffness, toughness and density are independent axes.

Beyond the linear law

$$U_{\text{rec}} = \tfrac{1}{2}\sigma\varepsilon, \qquad U_{\text{lost}} = \oint \sigma\, d\varepsilon$$

$$\text{fatigue}: \ \text{cracks grow below yield}$$

Past yield, dislocations move and deformation is permanent. Unloading follows a line parallel to the elastic slope, so only the elastic triangle returns; the rest is heat and permanent set. A hysteresis loop encloses the loss per cycle, which is how rubber damps and why a tuning fork rings.

WHY / Elastic aftereffect is slow partial recovery; elastic fatigue is weakening under repeated load. Each bend of a paperclip multiplies dislocations, raising the yield stress and shortening the life.

WATCH / After yield u = sigma^2/2Y no longer describes the material: the same load now leaves a permanent set.

Specific strength and specific modulus

$$\text{tie}: \ \frac{\sigma_y}{\rho} \text{ or } \frac{Y}{\rho}; \qquad \text{beam at fixed mass}: \ \frac{Y^{1/2}}{\rho}$$

$$\text{steel } 3.2\times10^{4},\ \text{Al } 7.4\times10^{4},\ \text{CFRP } 9.4\times10^{5}\ \text{m}^2\text{s}^{-2}$$

A cable of fixed breaking load weighs rho F/sigma_y, so the ranking is specific strength. A beam of fixed mass has I proportional to 1/rho^2 and deflection proportional to 1/YI, so the ranking is sqrt(Y)/rho. Different applications, different indices, possibly different winners.

WHY / Material selection is multi-objective. A composite wins both the cable and the bicycle frame here, but for two different reasons, and neither reason is simply the highest modulus.

WATCH / Comparing moduli alone always favours steel; comparing masses answers the engineering question. State the constraint before quoting a winner.

Buckling versus crushing

$$P_{cr} = \frac{\pi^2 YI}{L^2}, \qquad P_y = \sigma_y A$$

$$P_{\text{fail}} = \min(P_{cr}, P_y)$$

A short fat column crushes at a load set by strength. A long thin one bows sideways and fails at a load set by stiffness, proportional to I and to 1/L^2, with strength absent. A real column fails at whichever load is smaller.

WHY / Buckling is a geometric instability, not a material limit: the amplitude grows without any new stress, so the failure load does not depend on how strong the material is, only on how stiff and how slender.

WATCH / Comparing the two and keeping the larger is the trap. Slender columns buckle far below yield, and then the failure is sudden, with no warning from the stress-strain curve.

Fracture, fatigue and safety factors

$$n = \frac{\sigma_{\text{yield}}}{\sigma_{\text{working}}} \text{ or } \frac{P_{\text{fail}}}{P_{\text{working}}}$$

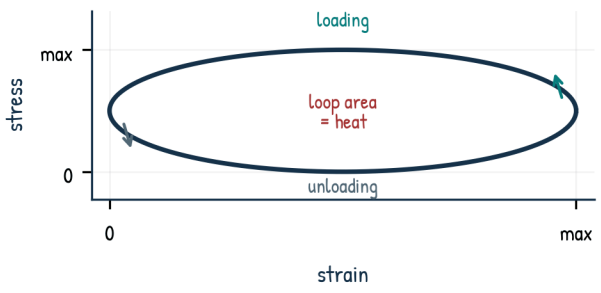
$$\text{brittle}: \ \text{no yield plateau, so no warning}$$

Toughness is the area under the stress-strain curve to fracture. Glass is stiffer than mild steel and far less tough: it snaps without warning. Fatigue cracks grow a little on each load cycle, so a member designed on a single static load can still fail in service.

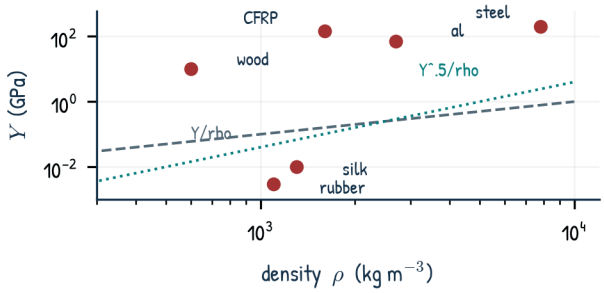
WHY / Safety factors cover three separate unknowns: scatter in material properties, uncertainty in the load, and the difference between a test specimen and a real joint with holes and welds.

WATCH / A safety factor does not repair a wrong model. If the mode is wrong, a larger factor only delays the failure it was never designed to prevent.

Hysteresis: unloading returns by a different curve; the loop is lost energy



Materials chart: stiffness and density are independent axes



REMEMBER / Stiffness is not strength, and neither one is toughness.

JEE toolkit: triage, traps and limit checks

Name the mode, then the section, then the load history, then the formula.

Five-step triage

$$\text{mode} \to \text{section} \to \text{load history} \to \text{formula} \to \text{limit check}$$

Is the body stretched, squeezed, sheared, twisted, bent or heated? Is it a wire, composite, shaft, beam or vessel? Is the load static, sudden or dropped? Only then choose among the ledger formulas, and finish with a limiting case.

WHY / The order exists because each step can be checked cheaply. A category error discovered at step one costs one line; discovered after the algebra, it costs the question.

WATCH / If two formulas both seem to apply, one is usually a limit of the other. Say which, and the ambiguity disappears.

The four different twos

$$\text{self-weight}: \ \tfrac{1}{2}mgL/AY$$

$$\text{sudden load}: \ 2\Delta_{\text{st}}$$

$$\text{hoop vs longitudinal}: \ 2:1$$

$$J = \pi R^4/2 \ \text{against} \ I = \pi R^4/4$$

Four unrelated factors of two appear in this chapter. Each has a one-line origin: a triangular integral, an energy root, a free-body cut of a cylinder, and a definition of the polar second moment.

WHY / Memorising "there is a factor of two somewhere" is not a method. If you can name the origin you can regenerate the number under exam pressure.

WATCH / The factor of two for a sudden load is a peak; the factor of two in the self-weight is an average. They are not interchangeable, and a problem can contain both.

Numbers and units worth carrying

$$Y_{\text{steel}} = 2.0 \times 10^{11}\ \text{Pa}, \quad G_{\text{steel}} = 8.0 \times 10^{10}\ \text{Pa}$$

$$\sigma_y \approx 2.5 \times 10^{8}\ \text{Pa}, \quad \alpha = 1.2 \times 10^{-5}\ \text{K}^{-1}, \quad \nu = 0.29$$

$$B_{\text{water}} = 2.2 \times 10^{9}\ \text{Pa}, \quad \rho_{\text{steel}} = 7800\ \text{kg m}^{-3}$$

These are the values used throughout the worked examples, rounded to two significant figures so the arithmetic stays checkable by hand.

WHY / A dimension check catches most slips: stress should land in 10^7–10^9 Pa, strain in 10^-4–10^-3, modulus in 10^10–10^11 Pa. An answer outside those ranges is a units or area error.

WATCH / Do not mix GPa and Pa inside one line. A stress written as 2 in GPa added to a strain will not announce itself; the final number is simply wrong by 10^9.

Scaling laws worth memorising

$$k \propto \frac{A}{L}: \ L \to 2L \Rightarrow \frac{k}{2}; \quad d \to 2d \Rightarrow 4k$$

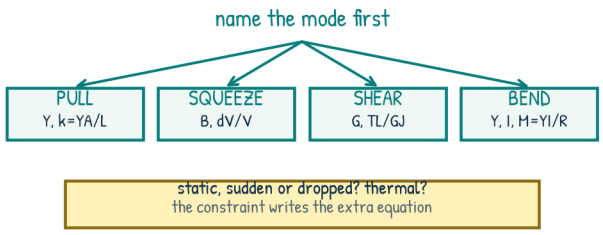
$$\theta \propto \frac{TL}{GR^4}; \qquad \delta \propto \frac{FL^3}{YI}$$

Stiffness falls with length and rises with area; twist falls as the fourth power of radius; deflection rises as the cube of length. Write each as a proportionality before substituting numbers, and the answer often needs no arithmetic.

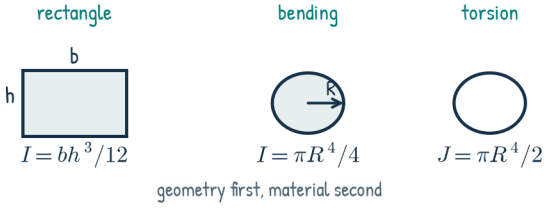
WHY / Scaling also ranks the variables. In torsion the radius enters as R^4 while the length enters to the first power, and in a beam the depth enters cubed while the width enters once.

WATCH / Scaling arguments assume the mode does not change. Make a shaft thin enough and it fails by wall buckling, not by twist, and the R^-4 law stops applying.

Triage: mode, then section, then load history, then the formula



The same circle has I = pi R^4/4 in bending and J = pi R^4/2 in torsion



REMEMBER / Most lost marks are category errors, not algebra errors.